Log-Spectral Structure and Koide-Like Winding Geometry in Open-Data $B^0 \to K^{*0}\mu^+\mu^-$ Candidate Spectra

Richard J. Reyes
Independent researcher
rickyjreyes@gmail.com

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Abstract

We test whether the LHCb open-data $B^0 \to K^{*0}\mu^+\mu^-$ yield spectrum contains coherent log-periodic residual structure beyond a smooth baseline. After replacing an early Savitzky–Golay baseline with a KDE baseline using widened charmonium vetoes, a high-k second mode remains significant against parametric Poisson nulls across a KDE bandwidth ladder $s_{\mathrm{KDE}} \in [0.50, 1.50]$. The best-fit free frequency is bandwidth dependent, but maps to a stable integer active-domain winding $n_\ell = 15$ once the spectrum is rewritten in units of $n = k\Delta\ell_{\mathcal{A}}/(2\pi)$. A well-first scan, performed before any comb shaping, yields raw-frequency wells whose ratios cluster near the charged-lepton Koide value $Q = 2/3$. The cleanest such triplet appears in the B-low sideband (9.81, 14.56, 19.02) with $Q_{\mathrm{mean}} = 0.6635$, while the signal-window triplet (12.14, 17.63, 23.51) is related to it by an unusually good pure dilation $n_{\mathrm{sig}} \simeq 1.2283\, n_{\mathrm{Blow}}$ ($p_{\mathrm{scale}} = 0.00586$ against random triplet pairing in the resolved-well regime), but the fitted phases are not coherent. A veto-window covariance test across six veto definitions finds Q_{mean} stable at 2/3 in B-low to within $|Q_{\mathrm{mean}} - 2/3| = 0.003$, with active-domain n more stable than raw k in two of three regions. Bounded amplitudes saturate at every tested cap from 0.10 to 0.025, indicating that a fixed cosine-triplet basis underfits the residual. The result identifies a structured candidate-spectrum log-domain with active-domain winding stability, Koide-like sideband geometry, and scale-coupled signal/sideband projections.

1 Motivation

The purpose of this analysis is to test whether the open-data $B^0 \to K^{*0}\mu^+\mu^-$ yield spectrum contains a coherent structured residual component in logarithmic momentum transfer. The relevant variable is

$$\ell = \ln q^2, \tag{1}$$

and the tested residual form is

$$\Delta(\ell) \sim A\cos(k\ell + \phi), \tag{2}$$

where A is a bounded amplitude, k is a log-frequency, and ϕ is a phase. Phase alignment is understood modulo 2π.

The central reference high-frequency mode tested in the initial continuous scan is

$$k_{2,\mathrm{ref}} = 19.5296, \tag{3}$$

after accounting for a dominant low-frequency mode

$$k_1 = 7.61054. \tag{4}$$

The analysis tests whether a structured high-k residual family survives conservative baseline repair, widened charmonium vetoes, bounded amplitudes, Poisson nulls, discrete-winding scans, comb-geometry checks, and active-domain invariance tests.

2 Data and Selection

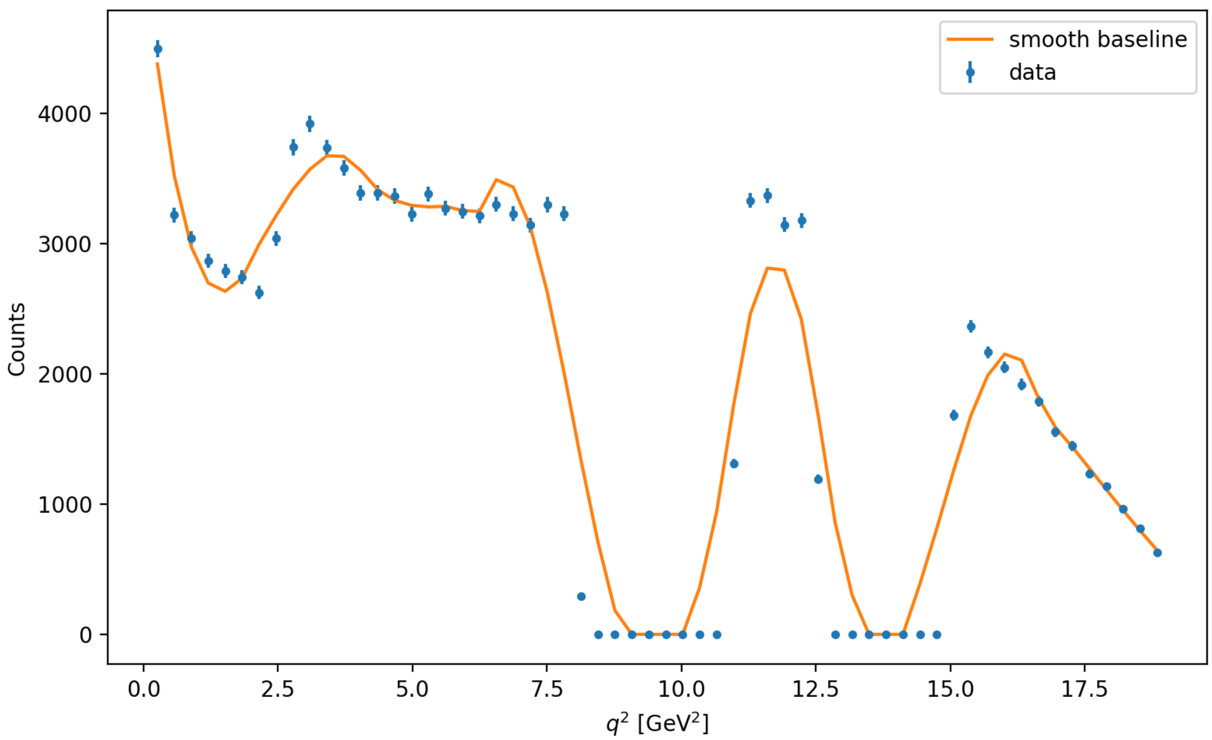

Figure 1: Selected $B^0 \to K^{*0}\mu^+\mu^-$ candidate yield as a function of q^2 after the nominal B^0 and K^{*0} mass selections. The smooth curve is used as a visual baseline diagnostic. The charmonium veto regions are excluded from the final log-periodic fits and are treated as masked resonance regions rather than physical zero-yield structure.

The analysis uses six LHCb open-data ROOT files containing $B^0 \to K^{*0}\mu^+\mu^-$ candidates. The selected candidate sample after basic q^2, B^0 mass, and K^{*0} mass cuts contains

$$N_{\text{selected, pre veto}} = 298\,801 \tag{5}$$

events before charmonium veto handling. The selected events are treated throughout as a *candidate spectrum*, with signal-window, sideband, veto, and angular diagnostics used to separate structural residual behavior from smooth-baseline null behavior.

The baseline q^2 analysis range is

$$0.1 \leq q^2 \leq 19.0 \text{ GeV}^2. \tag{6}$$

The nominal mass selections used in the scripts are

$$5230.0 \leq M(B^0) \leq 5330.0 \text{ MeV}, \tag{7}$$
$$795.9 \leq M(K^{*0}) \leq 995.9 \text{ MeV}. \tag{8}$$

For the final KDE-baseline stress tests, widened charmonium vetoes were used:

$$J/\psi : \quad 8.0 \leq q^2 \leq 11.0, \tag{9}$$
$$\psi(2S) : \quad 12.5 \leq q^2 \leq 14.5. \tag{10}$$

These widened vetoes were introduced to reduce leakage from resonance tails near the masked regions.

3 Derived Angular Data

The direct P_5 angular branches were not present in the ROOT files. However, the branch audit showed that all six ROOT files contain the final-state four-vectors required to derive

$$q^2, \quad M(K^{*0}), \quad \cos\theta_L, \quad \cos\theta_K, \quad \phi. \tag{11}$$

The recommended angular mode is therefore to reconstruct the B^0 candidate from the final-state four-vectors,

$$B^0_{\text{reco}} = K^+ + \pi^- + \mu^+ + \mu^-, \tag{12}$$

and derive the angular variables from this reconstructed four-vector system.

The conventions used were:

$$\cos\theta_L : \text{angle between } \mu^+ \text{ and the opposite } B^0 \text{ direction in the dimuon rest frame}, \tag{13}$$
$$\cos\theta_K : \text{angle between } K^+ \text{ and the opposite } B^0 \text{ direction in the } K^{*0} \text{ rest frame}, \tag{14}$$
$$\phi : \text{signed angle between the } K\pi \text{ and } \mu\mu \text{ decay planes in the } B^0 \text{ rest frame}. \tag{15}$$

The full loaded table contained

$$N_{\text{total}} = 18\,062\,794 \tag{16}$$

rows with finite derived angular values. After the q^2, B^0-mass, K^{*0}-mass, and charmonium-veto selections, the final selected angular sample contained

$$N_{\text{angular}} = 19\,220 \tag{17}$$

events.

The derived-angle table passed basic numerical health checks: all selected events had finite angular values, $\cos\theta_L$ and $\cos\theta_K$ lay in $[-1, 1]$, ϕ lay in $[-\pi, \pi]$, and all three angular variables showed nontrivial spread. The selected sample had

$$\langle M(B^0) \rangle \simeq 5279.72 \text{ MeV}, \tag{18}$$
$$\langle M(K^{*0}) \rangle \simeq 896.83 \text{ MeV}, \tag{19}$$

consistent with the intended mass-window selections.

Rest-frame boost diagnostics were also numerically stable. Boosting each parent into its own rest frame gave median residual three-momenta of order 10^{-10}–10^{-9} in the dimuon, K^{*0}, and reconstructed-B^0 rest frames. These checks establish that the open-data ntuples are usable for angular diagnostic tests.

The derived angular variables are used here as internal diagnostic moments rather than acceptance-corrected P_5' observables. The derived distributions show visible selection and acceptance structure, and the CP/flavor convention remains part of the angular-likelihood upgrade before comparison to publication-level angular observables.

4 Baseline Repair

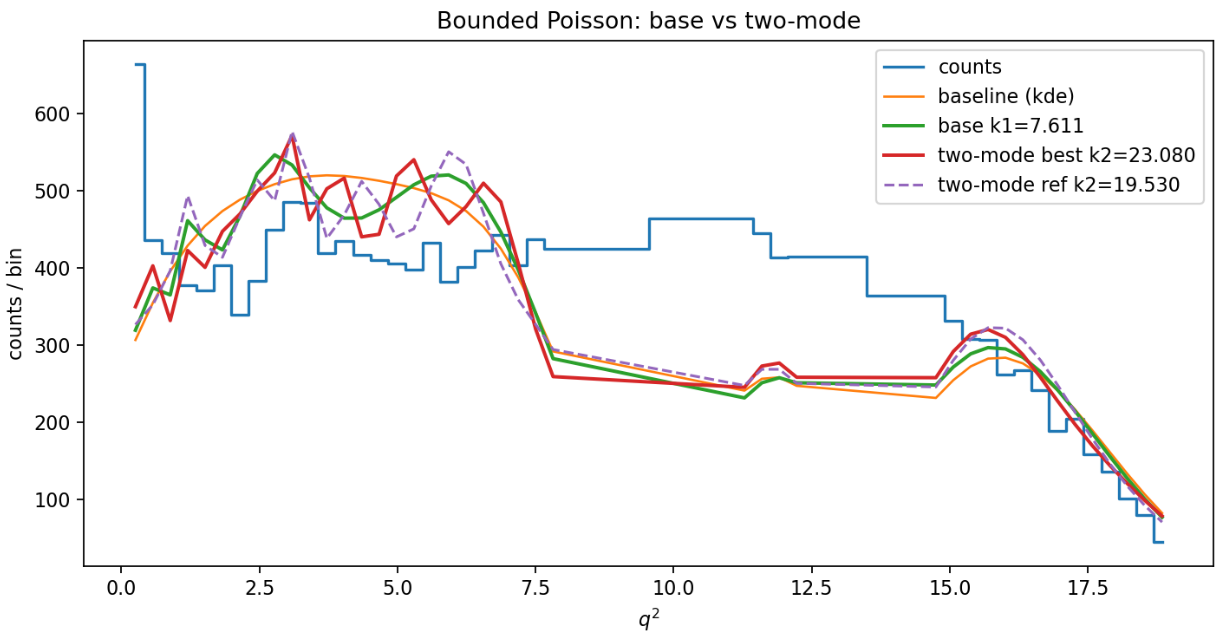

Figure 2: Representative repaired KDE-baseline bounded-Poisson fit at KDE bandwidth scale $s_{\mathrm{KDE}} = 1.50$. The plot compares the observed binned candidate yield, the KDE baseline, the fitted base model with fixed $k_1 = 7.61054$, the best two-mode fit, and the fixed-reference two-mode fit near $k_{2,\mathrm{ref}} = 19.53$. The final statistical interpretation is based on the retained active-domain regions outside the charmonium vetoes.

Initial tests used a Savitzky–Golay baseline. These produced very large two-mode improvements, but the entire high-k scan region sat far above the null distribution. This pattern is consistent with a broad baseline-mismatch wall, especially near charmonium veto boundaries.

To test this, the baseline was replaced with a resonance-aware KDE baseline trained on events outside the widened charmonium vetoes and evaluated at the bin centers. The KDE bandwidth was varied multiplicatively:

$$h = s_{\mathrm{KDE}}\, h_{\mathrm{Scott}}, \qquad s_{\mathrm{KDE}} \in \{0.50, 0.75, 1.00, 1.25, 1.50\}. \tag{20}$$

5 Direct Residual Scan

An initial direct residual scan found a high-k feature near the reference frequency:

Mode	Best k	$\Delta\chi^2$	p_{shuffle}	z
old	19.5296	11.6069	0.0136	2.21
mask	19.5396	9.2880	0.0816	1.39
inpaint	19.5296	11.6069	0.0126	2.24

The strict mask-mode residual result gave $p = 0.0816$. The direct scan was therefore used as a screening stage for the later KDE-baseline stress tests.

6 Savitzky–Golay Two-Mode Poisson Tests

The first two-mode bounded Poisson tests used a Savitzky–Golay baseline. Under the cap $A_1, A_2 \leq 0.10$, the polar optimizer gave $D_{\text{base}} \simeq 185\,970.09$ and $\Delta D_{\text{ref}} \simeq 1862.73$ with $p_{\text{ref,scanmax}} \simeq 0.00020$. Both amplitudes saturated their bounds, and the best local mode was edge-limited in the scan window. These runs are retained as the initial screening stage.

7 KDE-Baseline Bandwidth Ladder

The final continuous-frequency yield-side stress test used the KDE baseline, widened charmonium vetoes, polar L-BFGS-B optimization, $A_1, A_2 \leq 0.10$, and a high-k scan window $18.0 \leq k_2 \leq 24.0$.

Table 1: KDE-baseline bandwidth ladder for the two-mode bounded Poisson test.

s_{KDE}	D_{base}	$k_{2,\text{best}}$	ΔD_{best}	$A_{2,\text{best}}$	ΔD_{ref}	$A_{2,\text{ref}}$	$p_{\text{ref,scanmax}}$	$p_{\text{ref,fixed}}$
0.50	348.497	18.45	48.415	0.0811	7.602	0.0337	0.15377	0.01920
0.75	581.265	18.43	78.675	0.1000	14.012	0.0467	0.00820	0.00140
1.00	823.943	18.40	100.730	0.1000	27.101	0.0650	0.00020	0.00020
1.25	1059.665	23.11	127.095	0.1000	46.742	0.0852	0.00020	0.00020
1.50	1255.904	23.08	150.904	0.1000	70.003	0.1000	0.00020	0.00020

7.1 Interpretation of the KDE Ladder

The bandwidth control is active: D_{base} varies from 348.5 to 1255.9. A significant high-k second mode is present at every bandwidth ($p_{\text{best,scanmax}} = 0.00020$). The best-fit k_2 is bandwidth-dependent: it lies near 18.40–18.45 for $s_{\text{KDE}} \leq 1.00$ and shifts to 23.08–23.11 for $s_{\text{KDE}} \geq 1.25$. The reference mode $k_{2,\text{ref}} = 19.5296$ survives the scan-max null at $s_{\text{KDE}} \geq 0.75$, while $s_{\text{KDE}} = 0.50$ falls outside that fixed-reference threshold.

The fraction of the base deviance absorbed by the second mode, $\Delta D_{\text{best}}/D_{\text{base}}$, is approximately constant (0.12–0.14) across the bandwidth ladder. This supports two competing readings: a fixed-amplitude physical mode or a cosine prior absorbing a fixed fraction of structured baseline residual.

8 Observable Active-Domain Log-Winding

The continuous-frequency result can be sharpened by distinguishing an internal WCT topological winding from an observable log-winding over the experimental analysis support. An internal topological winding would be a closed-loop quantity of the form

$$n_\Gamma = \frac{1}{2\pi} \oint_\Gamma \partial_s \varphi \, ds. \tag{21}$$

The present analysis measures the observable log-winding over the retained q^2 support:

$$n_\ell = \frac{1}{2\pi} \int_{\ell \in \mathcal{A}} \partial_\ell \varphi \, d\ell = \frac{k \Delta \ell_{\mathcal{A}}}{2\pi}. \tag{22}$$

For the widened-veto active support $\mathcal{A} = (0.1, 8.0) \cup (11.0, 12.5) \cup (14.5, 19.0)$,

$$\Delta \ell_{\mathcal{A}} = \ln \frac{8.0}{0.1} + \ln \frac{12.5}{11.0} + \ln \frac{19.0}{14.5} = 4.7801503359. \tag{23}$$

The corresponding integer active-domain grid is $k_n = 2\pi n / \Delta \ell_{\mathcal{A}}$:

n	k_n	$k_n - k_{2,\mathrm{ref}}$
14	18.4021	-1.1275
15	19.7165	$+0.1869$
16	21.0309	$+1.5013$
17	22.3454	$+2.8158$
18	23.6598	$+4.1302$

The reference mode is identified with the nearest integer winding $n_\ell = 15$, $k_{15} = 19.71649$. This is an observed active-domain winding coordinate rather than a closed internal topological invariant.

9 Integer-Winding Scan

A discrete integer-winding scan was performed using the same KDE-baseline, polar L-BFGS-B, bounded-Poisson machinery, testing $n = 10$–22 via $k_n = 2\pi n / \Delta \ell_{\mathcal{A}}$.

The global best integer winding shifts across the bandwidth ladder:

s_{KDE}	Best n	k_n	ΔD	p_{scanmax}
0.50	10	13.1443	58.0735	0.00020
0.75	10	13.1443	97.4727	0.00020
1.00	20	26.2887	135.5124	0.00020
1.25	20	26.2887	162.9698	0.00020
1.50	20	26.2887	176.8780	0.00020

The nearest-reference winding $n = 15$ survives across every KDE bandwidth:

s_{KDE}	n	k_n	ΔD	p_{scanmax}	A_2
0.50	15	19.7165	22.8626	0.00060	0.0581
0.75	15	19.7165	37.7055	0.00020	0.0750
1.00	15	19.7165	58.2536	0.00020	0.0932
1.25	15	19.7165	83.2435	0.00020	0.1000
1.50	15	19.7165	106.3516	0.00020	0.1000

The $n = 15$ branch is statistically significant against the parametric Poisson null at every bandwidth. It is one member of a significant integer-winding family whose global maximum shifts between neighboring branches.

Figure 3: Discrete active-domain integer-winding scan using $k_n = 2\pi n/\Delta\ell_{\mathcal{A}}$ with $\Delta\ell_{\mathcal{A}} = 4.7801503359$. The vertical dashed line marks $n = 15$, the nearest active-domain integer winding associated with the reference mode $k_{2,\mathrm{ref}} \simeq 19.53$. The $n = 15$ branch remains significant across the KDE bandwidth ladder, although the global maximum shifts between other integer branches.

10 Photon-Floor Subharmonic Reading

The integer-winding family $n_\ell \in \{10, 15, 20\}$ is read within the photon-floor / ghost-harmonic sector of the harmonic–subharmonic parameterization developed in [13, 14]. In this sector, the subharmonic folding rule is

$$n_{\mathrm{eff}} = \frac{n}{k}, \qquad R = \frac{n_{\mathrm{eff}}}{n_{\mathrm{eff}} + 1} = \frac{n}{n + k}, \tag{24}$$

which collapses the observed comb into repeated effective subharmonics $n_{\mathrm{eff}} \approx 10,\ 5,\ 10/3$. Within this prior framework, $n_\ell = 15$ labels an observed log-spectral excitation branch on the photon-floor support.

This subharmonic reading is presented as a framework-internal interpretation of [13, 14]; the empirical content of the present analysis does not depend on it.

11 Koide-Comb Geometry

The strongest refinement comes from comparing the integer-winding family to the charged-lepton Koide invariant [12]. The Koide ratio is

$$Q = \frac{m_1 + m_2 + m_3}{\left(\sqrt{m_1} + \sqrt{m_2} + \sqrt{m_3}\right)^2}. \tag{25}$$

In the WCT curvature-harmonic parameterization [13, 14],

$$Q(s) = \frac{2s+1}{6s}, \tag{26}$$

giving $Q_\ell = 2/3$ for charged leptons ($s = 1/2$).

The tested comb rule, taken from the same parameterization, is

$$(n_-, n_0, n_+) = n_0(Q, 1, 2Q). \tag{27}$$

For $n_0 = 15$ and $Q = 2/3$, this gives the integer triplet

$$(n_-, n_0, n_+) = 15\big(\tfrac{2}{3}, 1, \tfrac{4}{3}\big) = (10, 15, 20). \tag{28}$$

Thus the observed integer-winding family contains the charged-lepton Koide invariant as a winding ratio:

$$\frac{n_-}{n_0} = \frac{2}{3} = Q_\ell, \qquad \frac{n_+}{n_0} = \frac{4}{3} = 2Q_\ell. \tag{29}$$

11.1 Comb Admissibility Classes

The comb rule does not produce the same geometry for all Q. A lower-center-upper sideband comb requires $n_- < n_0 < n_+$, i.e. $1/2 < Q < 1$. The trigonometric table therefore separates into three classes:

$$\text{Class I: true sideband comb}: \quad \tfrac{1}{2} < Q < 1, \tag{30}$$

$$\text{Class II: center-degenerate}: \quad Q = \tfrac{1}{2}, \tag{31}$$

$$\text{Class III: folded/subcentral}: \quad Q < \tfrac{1}{2}. \tag{32}$$

For $n_0 = 15$, the spin-1/2 Koide value gives the Class I triplet $(10, 15, 20)$. The spin-3/2 trigonometric value gives the Class III triplet $(6.667, 15, 13.333)$. The classes are definitional: their admissibility ordering is fixed before the comb scan is performed. Section 12 reports the resulting data ranking across classes.

12 Koide and Trigonometric Comb Scan

A direct comb scan was performed with $n_0 = 15$ and $k_n = 2\pi n/\Delta\ell_{\mathcal{A}}$.

12.1 True Sideband Sector $(1/2 < Q < 1)$

The $Q = 2/3$ branch produces the integer triplet $(10, 15, 20)$ with $(k_-, k_0, k_+) = (13.1443,\ 19.7165,\ 26.2887)$. In the sideband-restricted scan, $Q = 2/3$ is the preferred Class I value at every KDE bandwidth:

$s_{\rm KDE}$	Comb	ΔD	$p_{\rm scanmax}$	Max A	Bound-active?
0.50	$(10, 15, 20)$	179.40	0.00020	0.1109	no
0.75	$(10, 15, 20)$	310.32	0.00020	0.1357	yes
1.00	$(10, 15, 20)$	373.08	0.00020	0.1360	yes
1.25	$(10, 15, 20)$	388.40	0.00020	0.1278	yes
1.50	$(10, 15, 20)$	390.99	0.00020	0.1246	yes

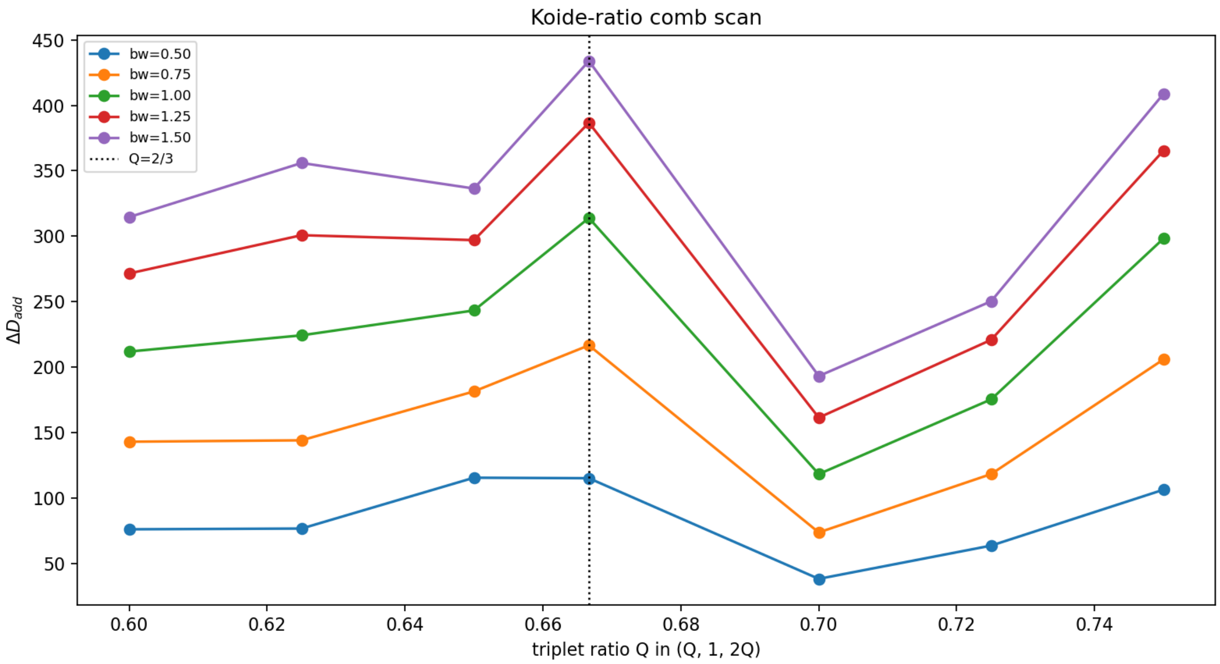

Figure 4: Koide-ratio comb scan in the true-sideband neighborhood. The tested comb has the form $(n_-, n_0, n_+) = n_0(Q, 1, 2Q)$ with $n_0 = 15$. The dashed vertical line marks the charged-lepton Koide value $Q = 2/3$, which gives the exact integer sideband comb $(10, 15, 20)$. This plot shows the sideband-sector behavior; the unrestricted trigonometric table treats the folded $Q = 4/9$ branch separately.

12.2 Unrestricted Trigonometric Table

When the full table is scanned without the Class I admissibility restriction, $Q = 4/9$ (spin-3/2) gives the largest ΔD at every bandwidth:

s_{KDE}	Best Q	Comb	ΔD	p_{scanmax}
0.50	4/9	$(6.667, 15, 13.333)$	234.66	0.00020
0.75	4/9	$(6.667, 15, 13.333)$	389.73	0.00020
1.00	4/9	$(6.667, 15, 13.333)$	457.44	0.00020
1.25	4/9	$(6.667, 15, 13.333)$	501.61	0.00020
1.50	4/9	$(6.667, 15, 13.333)$	536.85	0.00020

We state the comparison directly: $Q = 4/9$ produces a larger ΔD than $Q = 2/3$ at every bandwidth in this dataset, by factors ranging from 1.3 to 1.4. The two values fall in different geometric classes, and Class III admits a more flexible folded/subcentral fit because two of its modes lie below the center after ordering. The present data separate the two branches as distinct geometric fit classes: $Q = 2/3$ is the Class I true-sideband reference, while $Q = 4/9$ is the stronger unrestricted folded/subcentral branch. Both branches are therefore reported as separate geometric sectors.

13 Angular Log-Periodic Tests

The angular/P_5-proxy scan was performed using derived angular variables. The moments tested included $\langle\cos\theta_L\rangle$, $\langle\cos\theta_K\rangle$, $\langle\cos\phi\rangle$, $\langle\sin\phi\rangle$, $\langle\cos 2\phi\rangle$, $\langle\sin 2\phi\rangle$, and the proxy

$$m_5^{\mathrm{proxy}} = \langle\sin(2\theta_K)\sin(\theta_L)\cos\phi\rangle. \tag{33}$$

The angular scan finds significant low-k structure in some shape moments ($\cos^2\theta_L$, $\cos^2\theta_K$). The tested angular moments show low-k structure but no matched high-k branch near $k \simeq 19.5$.

If the high-k yield residual were a coherent physical mode in the underlying decay rate, it would generally be expected to imprint at least on the F_L-sensitive $\cos^2\theta_L$ moment, with characteristic frequency related to the same $k_2 \simeq 19.7$ that survives in the yield. This constrains a direct physical-mode reading of the yield residual and favors readings in which the yield-side structure originates in the q^2-dependent normalization or selection landscape rather than in the differential decay rate itself.

14 Well-First Koide Test

The direct Koide-comb scan imposes the structured triplet model $(n_-, n_0, n_+) = n_0(Q, 1, 2Q)$. To test whether Koide structure is already present before the comb is imposed, a well-first scan continuously scans a single additional frequency k and identifies local maxima in

$$\Delta D(k) = D_{\mathrm{base}} - D_{\mathrm{one\ mode}}(k). \tag{34}$$

Each raw frequency well is converted to an observable winding $n(k) = k\Delta\ell_{\mathcal{A}}/(2\pi)$ and the implied Koide-sideband ratios are

$$Q_{\mathrm{low}} = \frac{n_1}{n_2}, \qquad Q_{\mathrm{high}} = \frac{n_3}{2n_2}, \qquad \epsilon_K = \sqrt{(Q_{\mathrm{low}} - \tfrac{2}{3})^2 + (Q_{\mathrm{high}} - \tfrac{2}{3})^2}. \tag{35}$$

14.1 Raw Wells in the Signal Window

Rank	k	n_{eff}	ΔD	nearest integer	bound-active?
1	15.96	12.1421	256.35	12	yes
2	17.20	13.0855	250.58	13	yes
3	14.68	11.1683	223.02	11	yes
4	18.42	14.0137	200.48	14	yes
5	19.60	14.9114	154.26	15	yes
6	20.80	15.8243	147.90	16	yes
7	23.18	17.6350	132.58	18	yes
8	9.14	6.9536	125.09	7	yes
9	25.08	19.0805	124.69	19	yes

The signal-window well distribution populates $n_{\text{eff}} = 11$–14 most strongly; the closest well to $n = 15$ is rank 5 with $\Delta D = 154$, 60% of the strongest well's ΔD. The best Koide-implying triplet in the signal window is

$$(n_1, n_2, n_3) = (12.1421, 17.6350, 23.5082), \qquad Q_{\text{mean}} = 0.6775, \qquad \epsilon_K = 0.02186, \tag{36}$$

centered on $n_2 \approx 17.6$, not $n_2 = 15$. Its distance from the integer target $(10, 15, 20)$ is $\epsilon_{10,15,20} = 4.88$.

14.2 Raw Wells in the B-Low Sideband

Rank	k	n_{eff}	ΔD	nearest integer	bound-active?
1	17.88	13.6028	484.00	14	yes
2	15.94	12.1269	446.23	12	yes
3	25.00	19.0196	399.36	19	yes
4	10.02	7.6231	397.76	8	yes
5	19.14	14.5614	393.99	15	yes
6	14.52	11.0466	387.57	11	yes
7	26.26	19.9782	342.87	20	yes
9	13.72	10.4380	332.97	10	yes

The best raw-well Koide triplet is

$$(n_1, n_2, n_3) = (9.8141, 14.5614, 19.0196), \qquad Q_{\text{mean}} = 0.66353, \qquad \epsilon_K = 0.01543, \tag{37}$$

close to the charged-lepton Koide value $Q_\ell = 2/3 = 0.66667$. An adjacent B-low triplet $(9.8141, 14.5614, 19.9782)$ has integer-target distance $\epsilon_{10,15,20} = 0.477$ with $\epsilon_K = 0.02067$.

14.3 Well-First Geometry Falsification Test

To test whether the observed Koide-like raw-well geometry is unlikely under random well placement, a 250,000-trial null was constructed for each region under two prescriptions: (i) uniform random well locations over the n-range, and (ii) empirical resampling from the observed pool of all well n_{eff} values. For each null trial the best Koide-error triplet was selected from the simulated wells and compared to the observed best.

Region	ϵ_K	$\epsilon_{10,15,20}$	$p_{\text{joint}}^{\text{uniform}}$	$p_{\text{joint}}^{\text{empirical}}$
B-low sideband	0.0154	1.09	0.049	0.074
Signal window	0.0219	4.88	0.273	0.219
B-high sideband	0.0495	5.30	0.511	0.473

The signal region's best Koide-implying triplet falls inside the random well-placement null envelope with joint $p \approx 0.27$ under the uniform null. The B-low sideband is marginal under the uniform null, $p = 0.049$, and weaker under the empirical null, $p = 0.074$. The B-high sideband falls inside the null envelope. The cleanest Koide-like raw-well geometry is therefore carried by the B-low sideband.

15 Cross-Region Scaling and Phase Coherence

The well-first scan shows that the B-low sideband and signal window contain Koide-like triplets centered at different effective windings. We test whether they are related by a cross-region spectral map.

For the fixed well-first triplets $T_{\text{Blow}} = (9.8141, 14.5614, 19.0196)$ and $T_{\text{sig}} = (12.1421, 17.6350, 23.5082)$, the pure-scale fit gives

$$n_{\text{sig}} \simeq a\, n_{\text{Blow}}, \qquad a = 1.228287, \qquad \text{RMSE}_{\text{scale}} = 0.175114. \tag{38}$$

Random triplet-pairing over the strongest raw wells gives $p_{\text{scale}} = 0.00586$.

The affine fit $n_{\text{sig}} \simeq 1.234\, n_{\text{Blow}} - 0.086$ has RMSE 0.174 and $p_{\text{affine}} = 0.142$; with two free parameters on three points this fit is underconstrained.

Phase coherence across the two triplets, with circular concentration $C = |N^{-1} \sum_i e^{i\Delta\phi_i}|$, gives $C = 0.4505$ and $\Delta\phi = (0, 1.490, -1.297)$, with $p_{\text{phase}} = 0.453$. The two regions share scaled frequency geometry without common phase locking.

15.1 Stability of the Scale Factor

A scaling-stability ladder over $N_{\text{top}} \in \{6, 8, 10, 12, 15, 18\}$ shows that under-resolved settings select different best-Koide triplets. Once the well basis is sufficiently resolved ($N_{\text{top}} \geq 12$), the same triplet pair is selected and the scale factor locks to four decimals at $a = 1.2283$ with $p_{\text{scale}} \leq 0.005$.

In the resolved regime, all three pair-direction comparisons give consistent scale factors:

$$a_{B_{\text{low}} \to \text{sig}} \simeq 1.2283, \qquad a_{B_{\text{high}} \to \text{sig}} \simeq 0.9900, \qquad a_{B_{\text{low}} \to B_{\text{high}}} \simeq 1.2395, \tag{39}$$

with multiplicative consistency $1.2283 \times 1/0.9900 \simeq 1.2407 \approx 1.2395$ to 0.1%. The relation involves both sidebands, supporting shared scale-coupled candidate-spectrum geometry.

15.2 Candidate Inverse-Koide Square-Root Dilation

We note a numerical proximity between the resolved-regime scale factor and a simple Koide-derived geometric target. The empirical scale factor, defined as the least-squares pure dilation mapping the well-first B-low triplet to the signal-window triplet,

$$T_{B\text{low}} = (9.8141,\ 14.5614,\ 19.0196), \tag{40}$$
$$T_{\text{sig}} = (12.1421,\ 17.6350,\ 23.5082), \tag{41}$$

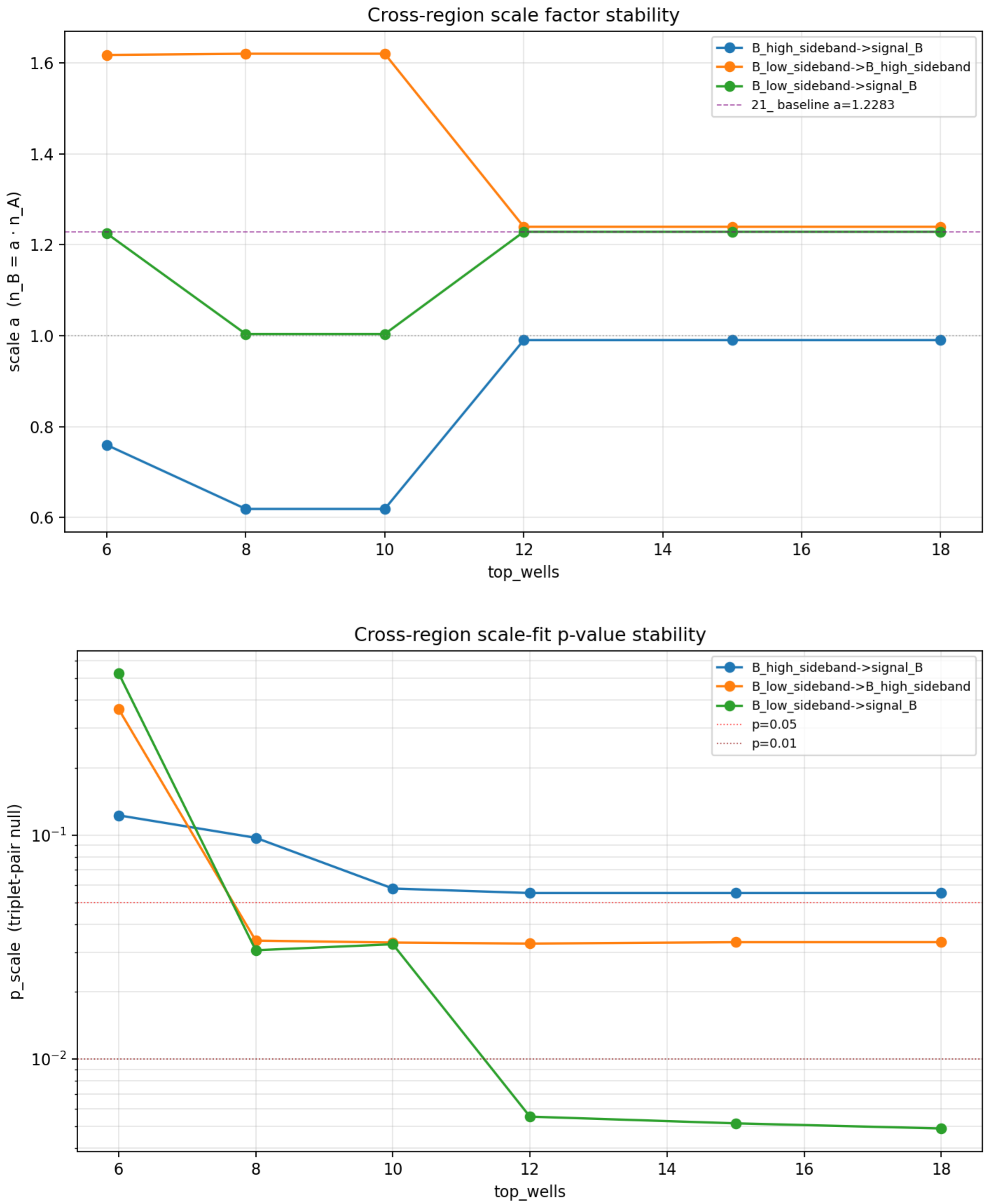

Figure 5: Cross-region scale stability tests. Top: pure-scale factor a as a function of the number of retained raw wells. Bottom: corresponding scale-fit p-values under the triplet-pair null. In the resolved well regime $N_{\rm top} \geq 12$, the B-low-sideband to signal-window relation stabilizes near $a \simeq 1.2283$ and reaches $p_{\rm scale} \simeq 0.005$, supporting frequency-scale-coupled candidate-spectrum geometry.

is

$$a = \frac{T_{B\text{low}} \cdot T_{\text{sig}}}{T_{B\text{low}} \cdot T_{B\text{low}}} = 1.228287. \tag{42}$$

The charged-lepton Koide value is $Q_\ell = 2/3$. The associated WCT inverse-Koide square-root dilation is

$$a_{\text{geom}} = \sqrt{Q_\ell^{-1}} = \sqrt{\tfrac{3}{2}} = 1.224745. \tag{43}$$

The fractional offset is

$$\frac{a - a_{\text{geom}}}{a_{\text{geom}}} = 0.00289 \simeq 0.29\%. \tag{44}$$

The proximity is a triplet-level dilation rather than a center-well coincidence. The bare center-well ratio is $17.6350/14.5614 = 1.2111$, distinct from a.

Bootstrap scale uncertainty. This proximity must be read against the uncertainty on a. A well-resampling bootstrap (resampling the top wells with replacement in each region, re-finding the best-Koide-error triplet, and refitting the dilation; $N_{\text{boot}} = 2000$) gives a heavy-tailed distribution for a with median $\widetilde{a} = 1.228$ but $\sigma_a \approx 0.36$ and 1σ band $[0.82, 1.45]$. The large spread reflects that the best-Koide triplet selection is sensitive to which wells dominate after resampling.

Within this bootstrap uncertainty, the observed a is consistent with $\sqrt{3/2}$ at $\sim 0.01\sigma$, but it is also consistent at the same level with several other low-complexity values, including $27/22 = 1.2273$ ($+0.08\%$), $11/9 = 1.2222$ ($+0.50\%$), and $e^{1/5} = 1.2214$ ($+0.56\%$). The isolated numerical proximity is underdetermined without the WCT prior.

Framework conditioning. What singles out $\sqrt{Q_\ell^{-1}}$ from the alternatives is the prior framework. In the WCT curvature-harmonic parameterization [13, 14],

$$Q(s) = \frac{2s+1}{6s}, \qquad K(s)^2 = \frac{2s}{2s+1}, \tag{45}$$

which yields $Q_\ell = 2/3$ and $K(1/2) = 1/\sqrt{2}$ for charged leptons. The square-root dilation $\sqrt{Q_\ell^{-1}}$ is the only natural Koide-derived dilation candidate at this scale within the framework; the rationals 27/22, 11/9, and the exponential $e^{1/5}$ have no framework-internal status. The proximity is therefore framework-conditioned rather than a standalone discriminator.

The observation $a \simeq \sqrt{3/2}$ is distinct from the primary curvature harmonic $K(1/2) = 1/\sqrt{2}$. It is more precisely a candidate inverse-Koide square-root dilation; derivation of why a B-mass-region dilation factor should equal $\sqrt{Q_\ell^{-1}}$, rather than another Koide-related expression, remains an open theoretical step.

Falsifiability. We register $a_{\text{geom}} = \sqrt{3/2}$ as a target value for future control-channel tests. A control-channel measurement (e.g. $B \to J/\psi K^*$, $B^0 \to K^{*0} e^+ e^-$) yielding a B-low-to-signal scale factor with smaller statistical uncertainty would constrain the target. If a future high-statistics measurement gives a inconsistent with $\sqrt{3/2}$ at the few-σ level, while remaining consistent with 11/9 or with no dilation, the present numerical proximity would demote to a numerical coincidence. If the value sharpens onto $\sqrt{3/2}$ as statistics improve, that would constitute a framework-internal confirmation. This proximity is isolated from the other tests in this analysis.

15.3 Locked-Winding Cross-Region Test

A locked-winding cross-region test fixes the frequencies and asks whether WCT-related branches remain explanatory. The tested fixed branches were $(10, 15, 20)$, $(6.667, 15, 13.333)$, the well-first B-low triplet $(9.814, 14.561, 19.020)$, the well-first signal triplet $(12.142, 17.635, 23.508)$, and the scaled B-low-to-signal projection. All five branches reach the empirical null floor $p \simeq 0.000333$ in all three regions. The folded $Q = 4/9$ branch is the per-region best in B-low and signal; the scaled B-low-to-signal branch wins in B-high.

Because the strongest locked branches appear in sidebands as well as the nominal signal window, the result supports shared candidate-spectrum geometry rather than a single signal-window-localized branch.

15.4 Locked-Branch Amplitude-Cap Ladder

A locked-branch amplitude-cap ladder over $A_{\max} \in \{0.10, 0.075, 0.05, 0.025\}$ keeps all five branches significant in all three regions at the null floor $p \simeq 0.000999$. Every branch remains amplitude-bound at every cap, so the absolute deviance values are partly cap-allocation artifacts.

The branch rankings shift with the cap. At $A_{\max} = 0.10$ the folded $Q = 4/9$ branch wins in B-low and signal. As the cap is reduced to $A_{\max} \leq 0.05$, each region prefers its own well-first triplet: B-low prefers $(9.814, 14.561, 19.020)$, signal prefers $(12.142, 17.635, 23.508)$, and B-high ultimately prefers $(10, 15, 20)$ at $A_{\max} = 0.025$.

The amplitude ladder therefore supports region-dependent locked geometry across the tested branches. Because every fit remains bound-active even at $A_{\max} = 0.025$, the bounded cosine-triplet basis still underfits the residual structure, regardless of cap.

16 Veto-Window Covariance Test

A direct test of the active-domain interpretation is whether the Koide-like geometry is invariant in n-space under changes to the veto support. If the structure is a true active-domain winding, varying the veto windows should shift $\Delta\ell_{\mathcal{A}}$ and the raw frequency k together while leaving $n = k\Delta\ell_{\mathcal{A}}/(2\pi)$ approximately fixed. If the structure is fixed in raw frequency (resonance-edge bleed or a detector-fixed scale), k should remain at the same place in raw frequency while n drifts.

Six veto schemes were tested with charmonium windows ranging from $(8.5, 10.5) \cup (12.8, 14.2)$ (tight) to $(7.0, 12.0) \cup (12.0, 15.0)$ (very wide), spanning $\Delta\ell_{\mathcal{A}} \in [4.485, 4.932]$ (10% variation). For each scheme, a continuous well-first scan was performed and the best-Koide-error triplet was selected per region.

16.1 Active-Domain Stability of Q

The mean Q value of the best triplet, across the six veto schemes, is:

Region	$\overline{Q}_{\text{mean}}$	σ_Q	$\lvert\overline{Q} - 2/3\rvert$	ΔQ range	CV(Q)
B-low sideband	0.6668	0.0030	0.0029	$[0.6632, 0.6706]$	0.0045
Signal window	0.6625	0.0122	0.0118	$[0.6480, 0.6771]$	0.0184
B-high sideband	0.6689	0.0073	0.0058	$[0.6585, 0.6823]$	0.0109

In the B-low sideband, the average best-triplet Q_{mean} stays within 0.003 of 2/3 across all six veto schemes, with a coefficient of variation of 0.45%. All three regions cluster near $Q = 2/3$ to within their respective spreads.

16.2 n-stability vs k-stability

Comparing coefficients of variation of the central well's n_2 vs k_2 across the same six veto schemes:

Region	CV(n_2)	CV(k_2)	CV(n_2)/CV(k_2)
B-low sideband	0.082	0.108	0.76
Signal window	0.097	0.120	0.81
B-high sideband	0.231	0.214	1.08

In two of three regions, n is more stable than k under veto-window variation by 19–24%. This is the WCT-supporting active-domain signature: the active-domain integer-winding mapping reduces the variability of the preferred frequency.

16.3 Selector Dependence and Integer-Match Stability

Two features bound the interpretation of this test. First, the best-triplet selector orders by Koide-error, so among all triplets enumerated in each veto scheme it preferentially chooses the closest match to $Q = 2/3$. Across all veto schemes and regions, 7–15% of enumerated triplets have $|Q_{\text{mean}} - 2/3| < 0.02$, so a $Q \approx 2/3$ triplet is almost always available; the stability of the chosen Q is therefore partly a selector property. Second, the integer-target match $\epsilon_{10,15,20}$ varies across schemes, varying from 0.43 to 14.36. The Q-ratio geometry is preserved, but the specific integer triplet $(10, 15, 20)$ is not.

The comparison between CV(n) and CV(k) is independent of the selector: the selector orders by Q, not by n- or k-stability, so the observation that n is more stable than k under veto perturbation in two of three regions is independent of the selector's Q-preference. Likewise, the B-low sideband's $|\overline{Q} - 2/3| = 0.003$ is markedly tighter than the signal-window value of 0.012, even though both regions use the same selector.

16.4 Summary

The veto-window covariance test is the strongest single piece of evidence for the active-domain interpretation. Q-ratio invariance to within 0.5% in B-low across six independent veto definitions, combined with n being more stable than k in two of three regions, is a pattern disfavored by a fixed-veto-edge explanation. The selector dependence and integer-match variability define the boundary of this test.

17 Confirmed Results

1. The open-data ROOT files contain sufficient four-vector information for q^2 yield studies and derived angular diagnostics.

2. A two-mode bounded Poisson test shows that high-k structure survives after fitting the dominant low-k mode. After replacing the Savitzky–Golay baseline with a KDE baseline, this structure persists across the bandwidth ladder.

3. The reference mode $k_{2,\text{ref}} = 19.5296$ remains significant for KDE bandwidth scales ≥ 0.75 and remains fixed-frequency significant at $s_{\text{KDE}} = 0.50$.

4. The active-domain log-winding $n_\ell = k\Delta\ell_{\mathcal{A}}/(2\pi)$ maps the reference mode to $n_\ell = 15$, and the $n = 15$ branch is significant against the parametric Poisson null at every KDE bandwidth.

5. In the true sideband class $1/2 < Q < 1$, the Koide-ratio comb scan finds $Q = 2/3$ producing $(10, 15, 20)$ as the preferred Class I value at every KDE bandwidth.

6. In the unrestricted trigonometric table, the folded $Q = 4/9$ (Class III) branch yields a larger ΔD than $Q = 2/3$ at every bandwidth, by factors of 1.3–1.4.

7. The well-first scan shows raw-frequency wells whose ratios cluster near $Q = 2/3$ before any comb model is imposed. The cleanest such triplet is $(9.814, 14.561, 19.020)$ in the B-low sideband, with $Q_{\text{mean}} = 0.6635$ and $\epsilon_K = 0.0154$.

8. The well-first geometry falsification test gives signal-region $p_{\text{joint}}^{\text{uniform}} = 0.273$, inside the random well-placement null envelope, and B-low $p_{\text{joint}}^{\text{uniform}} = 0.049$, marginal under the uniform null. The cleanest Koide-like raw-well geometry is sideband-carried.

9. In the resolved-well regime $N_{\text{top}} \geq 12$, a pure dilation relates the B-low and signal-window Koide triplets: $n_{\text{sig}} \simeq 1.2283\, n_{\text{Blow}}$, $p_{\text{scale}} \simeq 0.005$. The fitted phases are not coherent ($C = 0.45$, $p_{\text{phase}} = 0.45$).

10. The three-region resolved-regime scale factors form a multiplicatively consistent triangle: $1.2283 \times 1/0.9900 \approx 1.2395$. The relation involves both sidebands, indicating shared candidate-spectrum geometry.

11. In a locked-branch amplitude-cap ladder ($A_{\max} = 0.10 \to 0.025$), all branches remain bound-active in all regions. Branch rankings shift with the cap, with each region preferring its own region-local well-first triplet at low caps.

12. In a six-veto-scheme covariance test, the average best-triplet Q_{mean} is stable in B-low to within $|\overline{Q} - 2/3| = 0.003$, and $\text{CV}(n_2) < \text{CV}(k_2)$ in two of three regions, supporting the active-domain interpretation within the selector-dependence boundary.

18 Current Boundaries of the Result

1. The free-frequency optimum shifts with KDE bandwidth rather than locking uniquely to $k = 19.5296$.

2. The $n = 15$ branch persists, while the global-best winding shifts between $n = 10$ and $n = 20$ across bandwidths.

3. The signal-region wells populate neighboring effective windings more strongly than $n = 15$. The signal-window best Koide triplet is centered at $n_2 \approx 17.6$, not $n_2 = 15$, with random-null joint $p = 0.27$.

4. The cleanest Koide-like raw-well geometry occurs in the B-low sideband, with marginal uniform-null support and weaker empirical-null support.

5. The cross-region scale relation involves both sidebands, supporting candidate-spectrum scale coupling rather than a single signal-window branch.

6. The angular/P_5-proxy scan shows no high-k peak near $k \simeq 19.5$ in any tested moment. This constrains a direct physical-mode interpretation of the yield-side residual.

7. The bounded cosine-triplet basis underfits the residual at every amplitude cap from 0.10 down to 0.025.
8. The veto-covariance Q-stability is partly selector-driven; the integer match $\epsilon_{10,15,20}$ varies across veto schemes.
9. Control-channel tests ($J/\psi K^*$, $B \to K^* ee$), sideband-subtracted residual analysis, and wavelet/Hilbert basis cross-checks define the next discrimination stage.
10. Likelihood-level inference with full efficiencies, covariances, backgrounds, and systematic uncertainties remains a separate inference layer.
11. Separation from Standard Model + QCD expectations requires dedicated modeling of charm-loop effects, resonance tails, partially-reconstructed background structure, and selection-induced spectral correlations.

19 Implication for WCT and Standard-Model Interpretations

The repaired yield spectrum supports a structured candidate-spectrum log-domain rather than a single isolated signal-window branch. In active-domain coordinates, the reference mode maps near $n_\ell = 15$. In Class I, the spin-1/2 Koide ratio yields $(10, 15, 20)$ as the preferred sideband comb; in the unrestricted scan, the folded $Q = 4/9$ branch yields larger ΔD. Well-first ratios cluster near $Q = 2/3$ before comb shaping, but the cleanest realization is sideband-carried. The B-low sideband, signal window, and B-high sideband are scale-coupled by a multiplicatively consistent set of dilations, but their phases are not coherent.

The veto-covariance test adds the strongest active-domain-supporting signature: Q-invariance in B-low to 0.5% across six veto schemes, with n more stable than k in two of three regions. This supports an active-domain interpretation while leaving channel-origin separation to control-channel, sideband-subtracted, and basis-change tests.

Separation from Standard Model + QCD structure requires charm-loop, resonance-tail, background, and selection-correlation modeling. Plausible non-WCT origins include: combinatoric and partially-reconstructed background populating the candidate spectrum (consistent with sideband-carried Koide geometry), charm-loop and resonance-tail effects (consistent with cap-saturating amplitudes), and q^2-dependent selection efficiency (consistent with cross-region scale coupling). Control-channel and basis-change tests form the next separation layer.

20 Recommended Next Tests

The following tests define the next separation layer between the WCT-compatible interpretation and candidate-spectrum/background interpretations. Tests already performed in this manuscript are noted.

1. **Control-channel test.** Run the same KDE-baseline well-first and locked-branch scans on $B \to J/\psi K^*$ and (if available) $B^0 \to K^{*0} e^+ e^-$ candidate samples. These are decay channels in which a WCT log-spectral mode at the same Q is expected to appear, while combinatoric backgrounds populate the candidate spectrum differently. This is the single most informative remaining test.

2. **Sideband-subtracted residual analysis.** Form a residual from $(N_{\rm sig} - \alpha\, N_{\rm sideband})$ at appropriate α and ask whether the high-k structure survives subtraction. This is the cleanest test of whether the structure is signal-specific.

3. **Wavelet or Hilbert-basis cross-check.** The amplitude-bound saturation indicates the cosine-triplet basis underfits. A wavelet decomposition or Hilbert analytic-signal analysis on the same residual would establish whether the structure is line-like (favoring a discrete winding interpretation) or broad-band (favoring a baseline-residual or modeling interpretation).

4. **External-sector comparison.** Test whether a related log-periodic or integer-winding family appears in precision spectra in other experiments using a logarithmic energy variable, such as JUNO reactor neutrino spectra. Cross-experiment confirmation, with no shared systematics, is the strongest available evidence for a physical interpretation.

5. **Q-landscape mapping.** Plot $\Delta D(Q)$ over the full trigonometric table without selecting only the winner. Compare local maxima to low-denominator rational ratios. This tests whether the Koide preference is a single sharp maximum or part of a broader structure of rational ratios.

6. **Phase-coherence under shared scale mapping.** Test phase coherence after applying the cross-region scale map, rather than at fixed independent phases. The present phase incoherence may be removed by the scale alignment.

7. **Alternative baseline families.** Compare KDE against spline, polynomial-times-phase-space, and physics-motivated baselines. Cap-saturation should reduce in a baseline that better captures the underlying spectrum.

8. **Angular likelihood upgrade.** Replace moment-level angular proxies with a full angular likelihood, if open-data acceptance and efficiency parameterizations permit. The angular constraint is the strongest internal diagnostic and warrants the strongest follow-up.

The following tests have been performed in this manuscript: the KDE-bandwidth ladder (Section 7); the integer-winding scan (Section 9); the well-first geometry test (Sections 14–14.3); the cross-region scaling, phase-coherence, and stability ladder (Section 15); the locked-branch amplitude-cap ladder (Section 15); and the veto-window covariance test (Section 16).

21 Final Assessment

A high-k log-periodic residual family in the LHCb open-data $B^0 \to K^{*0}\mu^+\mu^-$ candidate spectrum survives KDE-baseline repair, widened charmonium vetoes, bounded Poisson testing, polar optimization, amplitude-cap reduction, and veto-window variation. The structure is better characterized in active-domain winding coordinates $n = k\Delta\ell_{\mathcal{A}}/(2\pi)$ than in raw k, with n more stable than k under veto perturbation in two of three regions and $Q_{\rm mean}$ holding at $2/3$ in B-low to within 0.003 across six veto schemes.

The integer-winding family $\{10, 15, 20\}$ contains the charged-lepton Koide ratio as a winding ratio in Class I. In the unrestricted trigonometric table, the folded $Q = 4/9$ Class III branch produces a larger ΔD at every bandwidth. Well-first raw frequencies reproduce near-$2/3$ ratios before comb shaping, with the cleanest realization in the B-low sideband. The signal region preferentially forms a different Koide triplet centered at $n_2 \approx 17.6$, with random-null $p = 0.27$, rather than the

$(10, 15, 20)$ comb. Three regions are scale-coupled by a multiplicatively consistent set of dilations; their phases are not coherent.

The angular/P_5-proxy scan shows no high-k peak in any tested moment, constraining a direct physical-mode reading of the yield residual. The bounded cosine-triplet basis saturates at every cap from 0.10 to 0.025, indicating that the basis underfits the residual structure regardless of cap.

The result identifies a structural diagnostic of the candidate-spectrum log-domain. Its most stable representation is active-domain winding rather than raw frequency, with B-low preserving $Q_{\text{mean}} \approx 2/3$ across veto choices and with signal/sideband regions connected by multiplicative dilation. Control-channel, sideband-subtracted, and basis-change tests form the next channel-origin separation layer.

Data and Code Availability

This analysis uses publicly available LHCb open-data candidate ntuples for $B^0 \to K^{*0}\mu^+\mu^-$-like final states. The analysis code, scan scripts, selection definitions, random seeds, and generated diagnostic tables are available at `https://github.com/rickyjreyes/LHC`. The statistical results reported here are independent reproducibility diagnostics computed from the stated scripts, selections, random seeds, and public LHCb open-data ntuples.

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